Practical Assignment: Simplify a Manufacturing Workflow

Role: Full Stack Developer
Maximum time: Five hours
Tasks: Working prototype + third-party integration plan

Use AI tools throughout the assignment. We want to understand how you research an unfamiliar business problem, make decisions, and turn them into a practical solution.

Task 1: Research and build a focused prototype

Suggested time: Three hours

You are building an ERP for a small Indian auto-parts manufacturer. The manufacturer sends components to an external vendor for heat treatment and manually reconciles dispatch records, vendor updates, quality reports, and invoices.

Build a clickable browser prototype that makes **partial receipts, quality inspection, and vendor discrepancies** easy to understand and resolve.

Scenario

A customer needs 1,000 finished components. Each component requires heat treatment before shipment.

- The manufacturer sends 1,000 components to a vendor.
- The vendor reports dispatching 800 components back.
- The warehouse physically receives 780.
- Quality inspection accepts 740 and rejects 40 for possible rework.
- The vendor invoices for treating 800 components.
- The customer shipment is due in two days.
- There is no other finished stock.

State your assumptions and preserve discrepancies rather than changing figures to make them agree.

What to build

Briefly research how two existing manufacturing ERP products handle this workflow. Include source links and explain what influenced your solution.

Your prototype should allow a production manager to:

- Understand current quantities, discrepancies, and customer delivery risk.
- Distinguish vendor-reported information from warehouse receipts and inspection results.

- Record an inspection or reconciliation action.
- See quantities, statuses, and next actions update consistently.
- Review a simple history of important actions.

Choose the screens and interactions yourself. Prioritize a complete, intuitive workflow with realistic content and attention to detail.

Simulate one external dependency

Use a mock vendor update or local JSON fixture to demonstrate importing a dispatch update. Importing the same update twice must not double-count quantities. Include a simulated failure and a retry action.

Clearly label simulations. No live accounts, production credentials, ERP installation, or production backend are required. Local persistence is sufficient.

Task 2: Create a third-party integration plan

Suggested time: Two hours

Map the third-party integrations you would consider when expanding this prototype into a manufacturing ERP.

Cover the areas below, add anything relevant that is missing, and mark items as essential, optional, or not applicable. Identify capabilities that should be built into the ERP rather than integrated separately.

Area	Coverage to investigate
Existing systems	Legacy ERP, accounting, purchasing, inventory and data migration
Customers and orders	CRM, customer portals, EDI, sales orders and forecasts
Suppliers and subcontractors	Purchase orders, vendor updates, dispatches, invoices and portals
Product engineering	CAD/PLM, bills of materials, drawings and revisions
Production	MES, scheduling and shop-floor reporting
Machines and maintenance	Industrial IoT, machine data, downtime and maintenance systems
Warehouse and logistics	Barcode/RFID, scanners, weighing scales, WMS, transporters and tracking

Quality	Inspection systems, laboratory results, certificates and traceability
Finance	Banking, payments and reconciliation
Statutory requirements	GST, e-invoicing and e-way bills, where applicable
People and access	HR/payroll, attendance, identity providers and SSO
Communication and documents	Email, WhatsApp/SMS, storage, OCR and electronic signatures
Reporting	BI, data warehouse and reporting exports

Keep your integration table to these three points:

Recommended approach	Fastest practical implementation	Failure and fallback
Suitable provider, open-source option, connector or protocol, with an official documentation link	How you would integrate it quickly using existing tools, APIs, webhooks, file exchange or reusable adapters	How you would handle outages, duplicates, conflicting data or unavailable APIs

Keep entries concise and group related integrations where appropriate. Flag unverified capabilities rather than assuming they exist.

Finish with a short action plan explaining:

- Your recommended application framework and integration tools.
- Which connectors, open-source projects or managed services would accelerate delivery.
- What you would reuse versus build yourself.
- Your implementation sequence and realistic deliverables for the first **two days, first week and first month**.

State your team-size and access assumptions. Account for dependencies such as credentials, sandbox access and vendor approvals.

Submission

Please share:

- Working prototype link or runnable project, source code and setup instructions.
- Short README with research links, assumptions and limitations.
- Integration table and action plan.
- Three-minute walkthrough covering the prototype and key integration decisions.

- AI tools used and one example of something you corrected or improved.
- Actual time spent on each task.

Stop at five hours, including documentation and recording. Identify anything unfinished. No paid tools, purchases, live transactions or external outreach are required.